

Andre C.A. de Carvalho

AI-Augmented Full-Stack Developer

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Full-Stack Engineer with 7+ years shipping web and mobile products, now focused on building and governing LLM systems in production — LangChain integration, hybrid RAG, prompt engineering, and cryptographic provenance for AI-generated artifacts (author of the SEIF protocol — seifprotocol.com). All on a solid product foundation: marketplace/payment integrations, performance, and maintainable architecture, with a strong data/BI background (ETL, SQL, Python).

- **AI / LLM:** LangChain · hybrid RAG · prompt engineering · MCP · Ollama
- **AI governance (SEIF):** Ed25519 provenance · auditable governance · transparency log
- **Marketplaces:** Mercado Livre · Amazon · Shopee
- **Payments:** PagarMe · PinPag
- **Microservices:** Go · gRPC · NATS · Kubernetes

HIGHLIGHT (AI)

SEIF — Cryptographic provenance & governance for AI

2026 – Present

Author · AI provenance & governance protocol · seifprotocol.com

- Designed a protocol for **cryptographic provenance + auditable governance** (sessions, cycles, classification) over AI-generated artifacts — Ed25519 signing with optional time-anchoring (OpenTimestamps).
- Built langchain-seif (Python package): a zero-modification **LangChain callback handler** that adds a governance plane — classification, quality-gating, provenance signing, and a transparency-log trail — over any chain, in local or hosted multi-tenant mode.
- Implemented **hybrid RAG** over the knowledge store (lexical + Ollama embeddings) with a cost-aware retrieval trigger; context compression from 50k+ raw tokens to ~3–5k.
- Authored the grounding / system-prompt layer and ran **empirical prompt-engineering experiments** (A/B across multiple models) on whether framing changes correctness — measured, not assumed.

Python TypeScript LangChain RAG Ollama Ed25519 OpenTimestamps MCP Next.js PostgreSQL

EXPERIENCE

Full-Stack Engineer

Jan 2025 – Present

Beezoo Labs — Multi-tenant SaaS platform for loyalty and digital commerce (malls & retail)

- Built from scratch the Showcase microservice (digital storefronts with QR codes, pricing policies, and catalog enrichment via NATS events) plus its mobile-first frontend — end-to-end feature: .proto → Go service → admin panel → public app.
- Led the app-whitelabel initiative in the Go microservices monorepo (gRPC, NATS JetStream, Outbox Pattern, PostgreSQL, KrakenD), enabling multi-tier mobile app configuration per tenant.
- Developed white-label Flutter apps for consumers and partners — loyalty, campaigns, Eventim integration, biometric login, AI batch for catalog, and unified design system.
- Solo-built an OCR service for Brazilian tax receipts (NF-e/NFC-e) in Python/gRPC with a hybrid cascade (QR → Tesseract → PaddleOCR → Vision LLM), raising accuracy from 28% to 75% on a 1,138 real-image dataset; deployed on Docker + Kubernetes.

Go TypeScript Dart Python gRPC Protobuf NATS PostgreSQL Next.js 15 React 19 Flutter Docker
Kubernetes Tilt

Hubsell — E-commerce Integration Platform

- Built frontend and integrations for Seller Center/Dashboard, connecting sellers to marketplaces and external APIs.
- Integrated payment gateways (PagarMe, PinPag) and evolved critical product flows.
- Shipped conversion-impacting features (POS and Shopping Assistant).
- Improved UX/performance on heavy screens using ElasticSearch + React Query.

React Next.js Node.js MongoDB ElasticSearch Zustand React Query Expo

Full-Stack Developer

2020 – 2022

3 UP Tech — 3Ponto Financial App

- Built and maintained web and mobile apps end-to-end.
- Backend development using a microservices architecture and legacy integrations.
- Led frontend initiatives: feature delivery, bug fixes, and performance improvements.

React React Native Node.js PostgreSQL Express Microservices

Freelance Consultant

2016 – 2019

Upwork — Data & Business Systems

- Built ETL pipelines and Power BI dashboards for help desk analytics.
- Implemented ERP workflows (Odoo/OpenERP) with custom business processes.
- Automated data models with SQL and Python for executive dashboards.

Python SQL PostgreSQL MySQL Odoo Power BI

SKILLS

AI / LLM

LLM integration (LangChain) · Hybrid RAG · Prompt engineering · MCP · Ollama · LLM-as-judge / quality gating · Ed25519 provenance

Frontend

React · Next.js · TypeScript · State (Zustand/Redux) · Storybook

Backend

Go · Node.js · Python · gRPC · REST APIs · NATS · Microservices

Data & Infra

PostgreSQL · MongoDB · Redis · Docker · Kubernetes · ElasticSearch · ETL/SQL

Quality

Jest · Vitest · Playwright · CI/CD

EDUCATION

- B.B.A. in Business Administration — [Institution, year]

PUBLICATIONS

- CARVALHO, A. C. A. de. **On the Unique Primitive Second-Order System with Golden-Ratio Damping: Mathematical Properties and Applications to AI Context Verification.** Zenodo, 2026. DOI: [10.5281/zenodo.19344678](https://doi.org/10.5281/zenodo.19344678).